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# Assessing the effects of green innovation, renewable energy, and environmental policy stringency on ecological and material footprints in G7 economies

Emna Omri<sup>1\*</sup> , Chiraz Feki<sup>1</sup> and Nouri Chtourou<sup>1</sup>

\*Correspondence:  
em.omri@gmail.com

<sup>1</sup> LED/Department of Economics,  
Faculty of Economics  
and Management of Sfax,  
University of Sfax, Sfax, Tunisia

## Abstract

Ecological pressure is a multidimensional measure that includes factors such as water and soil pollution, air quality, and raw material consumption. It is therefore important to consider the determinants of ecological stress using many indicators to account for all these dimensions. More precisely, the intention behind this investigation is to identify the impacts of green energy, green innovation, GDP per capita, and the rigor of environmental governance on ecological and material footprints, by analyzing data covering G7 countries over the 1990–2020 period. Therefore, the CS-ARDL panel approach is applied. Actually, the findings confirm that, in the long run, GDP per capita increases both ecological and material footprints by 2.819 and 21.154, respectively. Furthermore, the strictness of environmental policy and the use of renewable energies are key factors in lessening the ecological footprint (by  $-0.061$  and  $-0.038$ , respectively, in the long run). On the other hand, green innovation increases the material footprint (by 0.111 in the short run and 0.067 in the long run). Therefore, the resource intensity of green technologies must be considered. In fact, this research emphasizes the need for developed economies to adopt strategies that enable them to become not only more carbon-neutral but also less resource-intensive. Actually, these findings can help policymakers address the dual challenge of ecosystem preservation and resource efficiency by orienting their efforts toward circular and resource-efficient economic models.

**Keywords:** Ecological footprint, Material footprint, Environmental policy stringency, Renewable energies, Green innovation

## 1 Introduction

In fact, the modern world is increasingly concerned with attaining environmental sustainability despite accelerated economic expansion and rising energy needs. In this context, 17 Sustainable Development Goals (SDGs) have been endorsed to guide national policies worldwide by addressing environmental and economic challenges. Actually, economic growth, together with intensive energy use and excessive material extraction, has increased ecological pressure and slowed progress toward achieving the SDGs. In



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this context, examining the main drivers of the ecological footprint (EF) and the material footprint (MF) is an important step towards understanding how the SDGs can be reached.

The concept of EF was initially explained by Rees [49]. This concept is broader than the carbon footprint, as it presents an aggregation of ecological damages by tracking many activities, such as fishing grounds and forestland. In fact, it is a measure of the biocapacity required to satisfy human demand. Recently, this concept has been widely used in the literature to assess ecological damage [40, 52, 53]. However, this concept failed to account for the extraction and use of raw materials. The repercussions of human activities on the environment are commonly evaluated in many studies, namely those of Dalali et al. [18], Dahmani [16], Ghazouani [22], and Hassan et al. [24], through greenhouse gas emissions. However, it is of utmost importance to use more aggregated measures that reflect all aspects of ecological damage, including pressures on both ecosystems and raw material stocks.

Recently, governments have been implementing regulations and mitigation strategies to reduce the EF and domestic material consumption, driven by heightened awareness of environmental issues and the establishment of the SDGs. In fact, based on Porter and van der Linde [46], rigorous strategies stimulate innovation. Hence, many studies emphasize the importance of environmental policy stringency (EPS) in reducing ecological damage by fostering the use of clean technologies (e.g., [3, 29]). Moreover, the transition to renewable energies (RE) and the diffusion of environmentally related technologies are crucial measures for attaining sustainable development and guiding our modes of consumption and production towards cleaner ones (e.g., [60]).

Additionally, the EPS is strong in the G7 countries and is implemented to address market failures and achieve the SDGs. Environmental governance mechanisms include technology support measures and both non-market-based and market-based mechanisms. For example, carbon taxes belong to market-based instruments. In fact, France and Germany are the two countries with the most stringent environmental policies among the G7. For instance, France is considered one of the strictest countries, notably for its implementation of carbon taxes and subsidies to encourage the spread of renewable energy. According to the IEA [25], around 40% of global GDP and 25% of energy-related carbon emissions are produced by the G7 countries. For instance, in 2021, these nations set a challenging target to attain carbon neutrality by 2050. Therefore, to achieve this target, each country has set rigorous strategies to accelerate the diffusion of renewables and green technologies. Then, since the majority of emissions comes from the electricity sector, decarbonizing it is a priority. In this context, a fast transition to RE is a key measure to attain this objective.

The choice of the G7 economies stems from their history of developing green technologies and introducing supportive environmental strategies within stable economic environments. Therefore, studying the main drivers of sustainability in these nations could reduce uncertainties and accelerate the ecological transition in other countries. In fact, as leaders in setting environmental strategies, they can share lessons learned to reduce uncertainties and accelerate ecological transitions in other less developed countries, which can incorporate G7 guidelines and regulations and adapt them to their own circumstances. Even though a few studies, including those of Ahmad et al. [4],

Balsalobre-Lorente et al. [9], and Xia and Liu [59], have scrutinized the determinants of the EF in the G7, no study has yet inquired into the consequences of green innovation (GI), EPS, and RE on the EF. Additionally, this is the first study to consider the drivers of the MF in the G7. Indeed, the choice of these variables is not arbitrary, given that GI and RE present the basic elements for any green transition. Furthermore, without a strict environmental strategy, these measures will take time to become effective. Besides, assessing how RE, GI, and environmental governance influence both the material and ecological footprints is crucial for understanding the real sustainability of these developed countries. Unlike existing analyses, which rely on a single environmental indicator, this dual-footprint analysis affords a holistic assessment of whether the current green transition is accompanied by reductions in both carbon intensity and material dependency.

Considering the preceding paragraphs, the subsequent questions arise. Are RE and GI effective in decoupling the green transition from both ecological degradation and excessive material extraction? Moreover, to what extent are current environmental strategies successful in promoting low-carbon and resource-efficient economies?

Therefore, the objectives of this investigation can be formulated as follows.

- 1) To examine the short and long-term links between GI, RE, and EPS using material and ecological footprints.

- 2) To clarify the causal links between these key components of the green transition and the two footprint indicators.

Hence, the purpose of this investigation is to jointly explore the effects of EPS, RE, and GI on EF and MF in the G7. To achieve this, the CS-ARDL approach is adopted alongside a set of econometric tools. Specifically, a cross-sectional dependence test and two panel unit root tests are conducted. Finally, cointegration and Granger causality tests are applied.

The ensuing part of the paper is conceived in the following manner. Section 2 contains an assessment of the literature. The methodology and main outcomes are detailed in Sect. 3 and Sect. 4. The conclusion is detailed in Sect. 5.

## 2 Literature review

This review contains a summary of the literature on the links between GI, GDP per capita, strict environmental policies, RE, and the material and ecological footprints. Then, the hypotheses are formulated.

### 2.1 Studies linking EPS to ecological and material footprints

Sohag et al. [53] applied a CS-ARDL to demonstrate the relevance of environmental policies for controlling the EF in OECD nations. They focused on the predominant roles played by technological innovation and RE as channels between EPS and EF. In line with this research, Afshan et al. [3] confirmed the negative effect of EPS on EF across OECD economies. The same result was confirmed by Chu and Tran [13] for the same panel of countries. Furthermore, Balsalobre-Lorente et al. [8] found that strict environmental regulations are negatively linked to EF in APEC economies.

Moreover, Kongbuamai et al. [32] and Dai and Du [17] have described EPS as an important factor for improving sustainability by reducing the EF in BRICS and BRICST

countries, respectively. Recently, Kartal et al. [30] also investigated the case of BRICS countries. The main results showed that non-market-based environmental strategies are ineffective, however, the consequences of market-based strategies on EF differ across countries. In the case of G7 countries, Xia and Liu [59] demonstrated that EPS is effective in reducing EF across all quantiles. Additionally, Kartal et al. [29] found mixed effects of EPS on EF, which varied by country and quantile.

Concerning the MF, Karlilar and Pata [28] found that EPS negatively affects the MF in OECD countries. A similar finding was also validated by Wang et al. [57] for European countries. Along the same lines, Zhang and Xiang [61] demonstrated that EPS reduces MF across a panel of resource-rich countries.

The literature previously mentioned has been used to formulate the following hypotheses:

H1a: EPS reduces the EF.

H1b: EPS reduces the MF.

## 2.2 Studies linking RE to ecological and material footprints

Sethi et al. [52] studied the BRICS+nations using the nonlinear ARDL approach. The main outcomes confirmed that a positive shock to RE improves the environment by reducing the EF. However, a negative shock increases the EF. In line with this result, Liu et al. [35] proved that RE decreases EF in BRICS economies. In leading oil-consuming countries, Hasan et al. [23] validated the same mitigating influence of RE on EF. The same outcome was confirmed by Qing et al. [47] in the South Asian region. In a broader study spanning 130 countries with different income levels, Li et al. [34] highlighted the negative effect of RE on EF. In addition, they revealed that this effect is more pronounced in lower-income nations and is accentuated as RE consumption increases. Furthermore, Joof et al. [27] applied the bootstrap ARDL to elucidate the effects of RE, the housing market, and economic complexity on the EF in the USA. The predominant results confirmed that the consequence of green energy on EF is negative. Furthermore, Omri et al. [40] confirmed the attenuating effect of RE on both carbon emissions and EF in France by using a nonlinear ARDL.

Regarding the MF indicator, Regueiro-Ferreira and Alonso-Fernández [50] demonstrated that RE reduces the MF of fossil fuels. However, its effect on other MF categories is positive in European countries. Finally, Dada and Al-Faryan [15] confirmed the negative impact of RE on the MF in Saudi Arabia. Recently, Kumar et al. [33] confirmed the negative effect of RE on carbon footprint, MF and EF in G20 economies.

Hence, the following hypotheses are proposed:

H2a: RE has a mitigating repercussion on the EF.

H2b: RE has a mitigating repercussion on the MF.

## 2.3 Studies linking GI to ecological and material footprints

Javed et al. [26] and Ali et al. [5] brought out that GI is decreasing EF in Italy and the USA, respectively. Likewise, Ullah et al. [54] investigated the determinants of the EF in Pakistan. The main results demonstrated that GI, RE, and green growth are mitigating the EF. Moreover, Abbas et al. [1] scrutinized the effects of GI on the EF in resource-rich economies. Their findings confirm the negative impact of GI. Additionally, Nketiah

et al. [39] used the ARDL and NARDL approaches to determine the main drivers of the EF in Ghana. They demonstrated that green technology reduces the EF when applying the ARDL technique. Using the nonlinear ARDL technique, they found that a negative alteration in green technology had a negative influence on the EF. Saqib et al. [51] examined a group of countries with the largest ecological footprints and proved that green growth, RE use, and GI are crucial factors in reducing EF. Nevertheless, Bozatli and Akca [11] used the AMG estimator, along with other approaches, to determine the main contributors to the EF in OECD countries. They demonstrated that while the impacts of RE and environmental tax are negative, that of environmental technology is statistically non-significant.

On the other hand, Wang et al. [56] confirmed that green technology fosters the EF in BRICS economies. Similarly, Anu et al. [6] used quantile regression and OLS for a selected panel of G20 and G7 countries. They demonstrated a positive link between GI and EF. Finally, Aydin et al. [7] revealed that GI enhances the EF in Finland, Germany, and the Netherlands. Regarding the MF indicator, Karlilar and Pata [28] confirm the negative effect of GI on MF in OECD countries. Conversely, Ganda [21] demonstrated that GI has a positive effect on MF in the BRICS countries.

Based on the studies mentioned above, two assumptions are formulated, as follows:

H3a: GI has a reducing repercussion on the EF.

H3b: GI has a reducing repercussion on the MF.

## 2.4 Studies linking GDP per capita to ecological and material footprints

Mohamed et al. [37] analyzed the case of Malaysia and demonstrated that GDP per capita increases the EF. Likewise, Li et al. [34] revealed that per capita GDP increases the EF both across a panel of 130 nations and within three income groups. Moreover, Dai et al. (2023) revealed that economic expansion raises EF across a panel of six nations. Furthermore, Kihombo et al. [31] confirmed that economic growth, along with financial development and urbanization, is increasing pollution in West Asia and Middle East countries. Similarly, Quan et al. [48] demonstrated that energy intensity and economic expansion increase the EF in OECD economies. For the same panel, the EKC assumption for the MF is validated by Karlilar and Pata [28]. Additionally, Ulussever et al. [55] confirmed the positive impact of GDP on the MF in GCC countries. Finally, Pata and Karlilar [42] confirm the positive effect of GDP per capita on fossil-fuel-based MF across 24 OECD countries.

Based on the literature review, we state the following hypotheses:

H4a: GDP per capita increases the EF.

H4b: GDP per capita increases the MF.

## 3 Data and methods

The methodological approach pursued to investigate these interactions and test the hypotheses presented in this work is shown in this section. Variable descriptions are explained in Table 1. The data are collected from the Global Footprint Network (GFN), the Organization of Economic Cooperation and Development (OECD), and the World Development Indicators (WDI). This study investigates the case of the G7 and covers the

**Table 1** Variable descriptions and sources

| Variable                        | Symbol | Measure                                                            | Data base |
|---------------------------------|--------|--------------------------------------------------------------------|-----------|
| Ecological Footprint            | EF     | gha per person                                                     | GFN       |
| Material Footprint              | MF     | Tonnes per person                                                  | OECD      |
| Green Innovation                | GI     | Environment related technologies (inventions per person, millions) | OECD      |
| Renewable Energies              | RE     | Part of global energy                                              | WDI       |
| GDP per capita                  | GDP    | Constant 2015 US\$                                                 | WDI       |
| Environmental Policy Stringency | EPS    | 0–6 scale                                                          | OECD      |

period from 1990 to 2020. The study period is limited to 2020 because EPS data are not available after this date.

To analyze the determinants of the EF and the MF, two models are specified:

$$EF = f(EPS, GI, RE, \ln GDP) \quad (1)$$

$$MF = f(EPS, GI, RE, \ln GDP) \quad (2)$$

The two models can be rewritten with a time series and panel form specification as follows:

$$EF_{it} = \alpha_0 + \alpha_1 EPS_{it} + \alpha_2 GI_{it} + \alpha_3 RE_{it} + \alpha_4 \ln GDP_{it} + \mu_{it} \quad (3)$$

$$MF_{it} = \delta_0 + \delta_1 EPS_{it} + \delta_2 GI_{it} + \delta_3 RE_{it} + \delta_4 \ln GDP_{it} + \mu_{it} \quad (4)$$

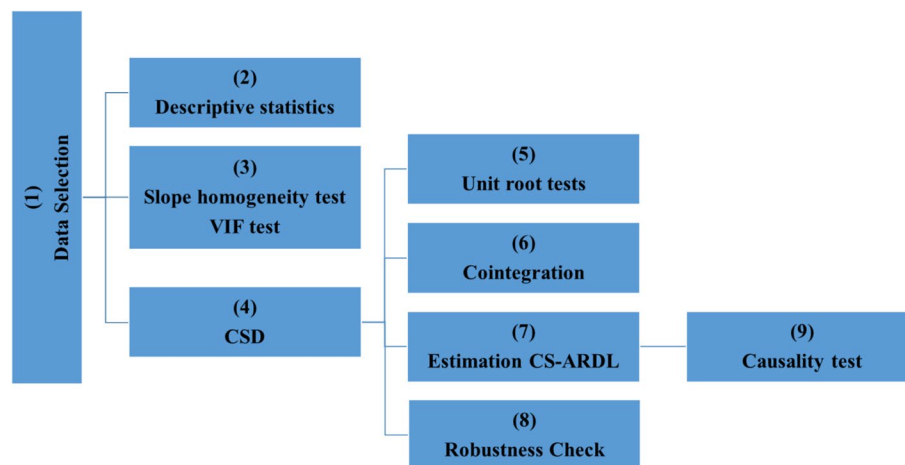
Because the G7 members have many common characteristics, the variables used are probably cross-sectionally reliant. In addition, it is highly likely that the G7 countries are exposed to the same macroeconomic shocks and share common policies and frameworks, and thus are not entirely independent. Hence, it would be very convenient to use the CS-ARDL, which explicitly stems from this dependence. In fact, unlike first-generation estimators or conventional panel ARDL, the CS-ARDL approach provides robust results in the presence of slope heterogeneity and cross-sectional dependence, while providing short and long-run estimates. This makes the CS-ARDL the most appropriate approach for this study's data and research objectives. Accordingly, the CS-ARDL estimation of Chudik and Pesaran [14] will be used.

$$Y_{it} = \sum_{i=0}^{P_{ef}} \beta_{i,t} Y_{i,t-1} + \sum_{i=0}^{P_x} \delta_{i,t} X_{i,t-1} + \sum_{i=0}^{P_z} \omega_i \bar{Z}_{t-1} + \mu_{i,t} \quad (5)$$

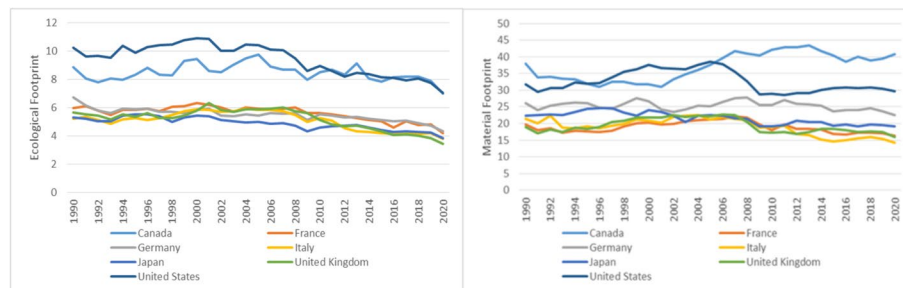
$$\bar{Z}_{t-1} = (\bar{Y}_{i,t-1}, \bar{X}_{i,t-1})$$

$Y_{it}$  is used to design endogenous variables, namely EF and MF,  $X_{i,t-1}$  denotes exogenous variables, namely GI, RE, EPS, and GDP per capita. Moreover,  $\bar{Z}_{t-1}$  denotes the average of all variables. Finally,  $P_{ef}$ ,  $P_x$ , and  $P_z$  denote each variable lag.

The adopted methodology contains several steps, as shown in Fig. 1. First, the descriptive statistics are detailed. Then, slope heterogeneity and the Variance Inflation Factor



**Fig. 1** Empirical approach



**Fig. 2** Time series plot of EF and MF across G7 countries (1990–2020)

(VIF) tests are applied. Next, the cross-sectional dependence (CSD) test of Pesaran [43] is used to assess CSD among the variables, which is frequent in macroeconomic data. After the CSD test, it is required to test the stationarity. Two unit root tests are chosen to tackle the problem of CSD and provide reliable stationarity results. Furthermore, the Westerlund cointegration test is applied to inspect for the occurrence of long-run interdependence among the variables. Once all these steps are completed, it is possible to proceed with the investigation of the links among the underlying variables using the CS-ARDL approach. Finally, the robustness check using the Dynamic Common Correlated Effects Mean Group (DCCMG) estimator and causality test are applied.

## 4 Empirical outcomes and discussion

### 4.1 Preliminary analysis

Figure 2 presents a time series plot of the two dependent variables, EF and MF, for the G7 over the period 1990–2020. Figure 2 illustrates a general decline in EF levels in all countries. This downward trend is more noticeable after 2008. Canada and the United States have the highest levels of EF. Although EF levels in both countries show a remarkable decline, they remain consistently higher than those in Japan and European countries. Particularly, Germany and the United Kingdom have noticeable downward trends. The EF in Japan maintains the lowest levels with a gradual



**Table 2** Descriptive statistics

|              | EF     | MF     | EPS   | GI     | RE     | lnGDP  |
|--------------|--------|--------|-------|--------|--------|--------|
| Mean         | 6.321  | 24.851 | 2.371 | 23.739 | 9.391  | 10.515 |
| Median       | 5.660  | 22.344 | 2.416 | 18.944 | 7.700  | 10.472 |
| Maximum      | 10.930 | 43.415 | 4.888 | 83.084 | 23.900 | 11.024 |
| Minimum      | 3.460  | 14.166 | 0.500 | 2.526  | 0.600  | 10.221 |
| Std. Dev     | 1.832  | 7.262  | 1.050 | 19.340 | 6.607  | 0.184  |
| Observations | 217    | 217    | 217   | 217    | 217    | 217    |

**Table 3** VIF and slope homogeneity tests

| VIF test                   |           |       |       |           |
|----------------------------|-----------|-------|-------|-----------|
| Statistics                 | EPS       | GI    | RE    | lnGDP     |
| VIF                        | 1.54      | 1.45  | 1.13  | 1.04      |
| 1/VIF                      | 0.650     | 0.689 | 0.886 | 0.963     |
| Mean VIF                   | 1.29      |       |       |           |
| Slope homogeneity test     |           |       |       |           |
|                            | Model 1   |       |       | Model 2   |
| $\Delta^{\sim}$ test       | 11.234*** |       |       | 12.704*** |
| $\Delta^{\sim}_{adj}$ test | 12.509*** |       |       | 14.147*** |
| ***p < 0.01                |           |       |       |           |

\*\*\*p &lt; 0.01

reduction over time. Concerning the MF, Canada and the United States maintain the highest levels. While Canada shows an upward trend, the United States exhibits a decline after the mid-2000s, followed by a period of stabilization. Japan and European countries exhibit relatively low and stable MF levels.

Table 2 shows a brief statistical analysis of the analyzed variables. According to this table, the highest median value (22.344) is associated with MF, and EPS has the lowest median value (2.416). Regarding standard deviation, lnGDP has the lowest volatility, while GI has the highest. Finally, the lowest mean value (2.371) is associated with EPS, while MF has the highest mean value (24.851).

The findings of the multi-collinearity test in Table 3 indicate that VIF and 1/VIF values for all independent variables are below 5 and above 0.2, respectively. These results confirm the absence of multi-collinearity in the two models. Concerning the potential presence of slope heterogeneity, the test of Pesaran and Yamagata [45] is adopted. The main finding in Table 3 contradicts the null assumption that slopes are homogeneous, indicating that coefficients differ across G7 countries.

The G7 economies share many similar economic and environmental characteristics, which potentially lead to interconnection among those countries. Consequently, CSD may be present in the dataset. For this reason, our analysis includes the CSD test of Pesaran [43]. The results in Table 4 indicate the occurrence of CSD. Since it has already been confirmed that the variables exhibit CSD, it is highly recommended to apply unit root tests of second-generation. Hence, we apply the CADF and CIPS tests of Pesaran [44]. The results presented in Table 4 point out that the integration



**Table 4** CSD and unit root tests

| Variables | CSD      | CIPS        |            | CADF       |            |
|-----------|----------|-------------|------------|------------|------------|
|           |          | I(0)        | I(1)       | I(0)       | I(1)       |
| EF        | 19.48*** | − 3.294 *** | − 5.646*** | − 2.031**  | − 7.539*** |
| MF        | 7.55***  | − 1.350     | − 4.392*** | 0.153      | − 4.515*** |
| EPS       | 23.30*** | − 2.550 **  | − 5.750*** | − 2.481*** | − 5.952*** |
| GI        | 24.66*** | − 2.455 **  | − 4.682*** | − 1.765**  | − 5.161*** |
| RE        | 19.77*** | − 0.683     | − 5.142*** | 3.996      | − 4.953*** |
| lnGDP     | 22.08*** | − 1.361     | − 4.259*** | 1.780      | − 5.312*** |

\*\*\*p &lt; 0.01, \*\*p &lt; 0.05

**Table 5** Outcomes of the cointegration test

| Statistics | Value      | Z-value | P-value |
|------------|------------|---------|---------|
| Model 1    |            |         |         |
| Gt         | − 3.323*** | − 3.459 | 0.000   |
| Ga         | − 7.393    | 0.925   | 0.823   |
| Pt         | − 7.101**  | − 2.229 | 0.013   |
| Pa         | − 6.932    | − 0.296 | 0.384   |
| Model 2    |            |         |         |
| Gt         | − 2.727    | − 0.782 | 0.217   |
| Ga         | − 4.595    | 2.887   | 0.998   |
| Pt         | − 8.002**  | − 2.140 | 0.016   |
| Pa         | − 7.772    | 0.535   | 0.704   |

\*\*\*p &lt; 0.01, \*\*p &lt; 0.05

orders of all variables are either I(1) or I(0), allowing the application of the CS-ARDL approach.

The cointegration of the two Models is verified by applying Westerlund's [58] cointegration procedure. The results in Table 5 indicate that EF and MF are associated with their determinants in the long run under both modeling frameworks. The existence of a long-run relationship across the two models at the panel level is verified, as the Pt statistic is significant. More precisely, in model 1, the Pt and Gt statistics are both significant, indicating that the long-run relationship exists not only at the overall panel level but also for at least one country. Concerning Model 2, cointegration is confirmed at the panel level, however the long-run relationship may not be observed for every country.

#### 4.2 CS-ARDL results

Table 6 reports the results of the panel CS-ARDL approach. The negative and significant coefficients of the ECT indicate that the two models adjust to the short-run disequilibrium by approximately 94.1% and 96.5% in the next period.

The coefficients in Table 6 reveal that the EF and MF are strongly and positively affected by GDP per capita. This finding is consistent with hypotheses H4a and H4b. This reveals that whenever GDP per capita rises, the EF and MF will persistently increase, leading to a deterioration in environmental quality. This outcome aligns with the empirical findings of Ahmad et al. [4] for the G7 and Ma et al. [36] for a panel of selected

**Table 6** The CS-ARDL results

| Variable               | Model 1     |         | Model 2     |         |
|------------------------|-------------|---------|-------------|---------|
|                        | Coefficient | P value | Coefficient | P value |
| Short-run coefficients |             |         |             |         |
| ECT                    | −0.941***   | 0.000   | −0.965***   | 0.000   |
| RE                     | −0.070*     | 0.063   | −0.226      | 0.440   |
| GI                     | 0.012       | 0.245   | 0.111*      | 0.053   |
| lnGDP                  | 5.765***    | 0.006   | 35.106**    | 0.017   |
| EPS                    | −0.124***   | 0.002   | 0.147       | 0.811   |
| Long-run coefficients  |             |         |             |         |
| RE                     | −0.038**    | 0.045   | −0.147      | 0.342   |
| GI                     | 0.005       | 0.333   | 0.067**     | 0.044   |
| lnGDP                  | 2.819***    | 0.004   | 21.154**    | 0.025   |
| EPS                    | −0.061***   | 0.001   | 0.026       | 0.939   |

\*p &lt; 0.1, \*\*p &lt; 0.05, \*\*\*p &lt; 0.01

economies. This result is likely due to the fact that higher GDP per capita is accompanied by greater consumption. This increase in consumption puts more pressure on these countries' natural resources, particularly energy resources. To meet this consumption, there will be more energy and raw material use, and more polluting emissions, which will increase their EF and MF. It is therefore advised to reinforce the stringency of environmental policies in these countries, not only those aimed at reducing CO<sub>2</sub> emissions, but also those that encourage embracing change in clean production and consumption methods to be resource-efficient and in harmony with the circular economy.

While the positive impacts of GDP on EF and MF is clear, a closer analysis shows that long-run coefficients (2.819 for EF and 21.154 for MF) are significantly smaller than short-run ones (5.765 for EF and 35.106 for MF). This implies that G7 economies respond to growth in a resource-intensive way in the short run. On the other hand, in the long run, structural adjustment and decoupling mechanisms alleviate this effect through energy efficiency, technological progress, and environmental governance. These findings reveal relative decoupling: GDP per capita increases EF and MF, but at a decreasing rate in the long run. For instance, G7 countries have shown a great interest in digitalization of the economy, which has played a key role in decoupling GDP growth from material-intensive activities. In addition, the structural transformations made in the energy sector (e.g., renewable energy in Germany, nuclear-based electricity in France, coal phase-out in the United Kingdom) have gradually reduced emissions from the energy sector and consumption of raw materials. Finally, Japan stands out for its circular manufacturing practices and high recycling rates, which have been reflected in low MF in the long run.

Furthermore, short and long-run findings confirm that an increase in EPS decreases the EF, but has no significant effect on the MF. This finding supports the initial hypothesis H1a concerning the EF. Hence, EPS is playing a decisive role in protecting the environment. These findings are supported by Ma et al. [36] and Chen et al. [12] in a panel of OECD and non-OECD nations, and by Xia and Liu [59] in G7 countries. This result is explained by the fact that environmental policies are essentially composed of carbon taxes and quotas on polluting emissions, which will reduce the consumption of fossil

fuels and air pollution, thereby automatically reducing the EF. Hence, stricter environmental strategies decrease the EF through imposing taxes on polluting activities and facilitating the adoption of environmentally friendly technologies. In fact, the environmental policies are more oriented toward pollution reduction than the efficient use of raw materials. It is therefore recommended that the EPS index include the dimension concerning resource efficiency and the circular economy. A closer look at the short and long-run impacts of EPS on EF reveals that stricter environmental regulations engender a short-run decrease in EF due to rapid behavioral responses from firms and households ( $-0.124$ ), but this effect diminishes in the long-run ( $-0.061$ ). This finding shows that EPS is effective at rapidly reducing the EF. To sustain this beneficial effect in the long run, stringent environmental policies alone are insufficient, and structural transformations are required.

Moreover, short and long-run results confirm that RE reduces the EF, but has no significant effect on MF. Hence, it is evident that the use of RE is improving not only air quality by reducing carbon emissions but also reducing the EF. This result supports hypothesis H2a, which states that the use of green energy has beneficial effects on the EF. These outcomes are consistent with those reported by Liu et al. [35] and Omri et al. [40]. Hence, RE is an efficient factor that enables the improvement of the ecosystem, its diffusion triggers an immediate effect on the EF. These results might be linked to the fact that enhancing the share of green energies in the energy mix by gradually replacing coal, natural gas, and oil helps reduce carbon dioxide emissions, a significant component of the EF. Hence, the gradual substitution of fossil fuels by RE enables reductions in environmental externalities, thereby lowering the EF.

In the first model, the GI has no significant effect on the EF, however its impact on the MF is positive in Model 2. Hence, the enhancement of sustainable technologies is increasing the MF. Destek and Manga [19], who proved that the effect of technological innovation on the EF is non-significant, further support this result. This result also aligns with Ganda [21], who demonstrated that GI has a positive effect on MF in BRICS countries. However, this result contrasts with Anu et al. [6] for a selected panel of G20 and G7 countries and with Aydin et al. [7] for European Union countries, who demonstrated that GI increases the EF. This result contrasts with that of Karlilar and Pata [28], who confirm the negative effect of GI on the MF in OECD countries. Our results thus indicate that GI does not have a significant effect on ecological pressure, however, its widespread adoption increases the consumption of raw materials. This results from the fact that green technologies are material-intensive and require increasing amounts of mineral resources and rare metals for the manufacture of photovoltaic panels, wind turbines, and batteries, including aluminum, lithium, cobalt, copper, etc.

Furthermore, the results confirm that GI has a smaller positive long-run coefficient (0.067) than the short-run one (0.111). This finding, in addition to emphasizing the material-intensive aspect of green technological innovation, reveals the presence of learning-by-doing in the long run. In fact, it is well known that green technologies such as wind turbines, solar panels, and batteries require large quantities of lithium, copper, and steel. In addition, the required infrastructure, such as electric vehicle charging, grids, and storage, is highly material-intensive. Due to this transition-phase effect, the short-run coefficient is very pronounced. In the long run, recycling systems, learning

effects, and supply chains are optimized, which weakens the positive effect over time. For instance, electric vehicles are more material-intensive than conventional vehicles. In fact, the expansion of battery manufacturing required to respond to the rapid diffusion of electric vehicles triggers a rapid increase in demand for critical minerals. While in the short run the focus is rather on performance and development, the material minimization and recycling systems gain a foothold over time, which explains the smaller long-run effect. Finally, this result could be explained by the rebound effect. In fact, improvements in the efficiency of renewable energy technologies and electric vehicles made them cheaper, thereby increasing total demand. This fact leads to greater total resource extraction and, obviously, an increase in the MF. At the macroeconomic level, environmentally oriented technologies reduce the costs of production and renewable energy systems, through increased efficiency. A decrease in costs stimulates consumption, investment, and growth, which increases total material demand.

#### 4.3 Robustness check and causality results

To strengthen the validity of our findings, a robustness check using the DCCMG estimator is applied, as presented in Table 7. This phase assures that the CS-ARDL results are reliable. Indeed, the DCCMG outcomes support the initial long-run results of the CS-ARDL model. In particular, GDP per capita is a significant driver of both ecological and material footprints, confirming that their economic models are not sustainable and continue to rely heavily on material extraction. Furthermore, the results confirm the importance of RE and EPS in reducing the EF by directly affecting carbon emissions, however their importance in reducing the consumption of raw materials is not significant. Finally, the expansion of green technologies increases the demand for raw materials.

Once long-term cointegration is verified and the coefficients are estimated using the CS-ARDL approach, the causal interlinkages among the variables are examined in Table 8. The results of the causality test aim to assess the potential causal links between the material and ecological footprints and the independent variables, thereby providing a deeper comprehension of environmental dynamics in the G7 countries. The results of the Granger causality approach of Dumitrescu and Hurlin [20] demonstrate the existence of causality running from all the determinants to both EF and MF. Moreover, there are two-way causal links between GI and EF, GI and MF, and finally between RE and MF. For instance, an increase in MF is generally accompanied by an increase in costs, greater dependence on some regions, and, consequently geopolitical tensions. This fact

**Table 7** Robustness check with DCCMG

| Variables | Model 1   |         | Model 2  |         |
|-----------|-----------|---------|----------|---------|
|           | Coef      | P-value | Coef     | P-value |
| RE        | −0.192*** | 0.000   | −0.397   | 0.258   |
| GI        | 0.051     | 0.235   | 0.118*   | 0.055   |
| lnGDP     | 6.535*    | 0.084   | 36.554** | 0.015   |
| EPS       | −0.272**  | 0.022   | 0.065    | 0.909   |

\*\*\*p < 0.01, \*\*p < 0.05, \*p < 0.1

**Table 8** Causality results

| EF is the dependent variable |                   |                        | MF is the dependent variable |                   |                        |
|------------------------------|-------------------|------------------------|------------------------------|-------------------|------------------------|
| Baseline assumption          | Zbar-stat         | Causality              | Baseline assumption          | Zbar-stat         | Causality              |
| EF $\nrightarrow$ EPS        | −0.382 (0.702)    |                        | MF $\nrightarrow$ EPS        | −0.655 (0.512)    |                        |
| EPS $\nrightarrow$ EF        | 6.026*** (0.000)  | EPS $\rightarrow$ EF   | EPS $\nrightarrow$ MF        | 8.236 *** (0.000) | EPS $\rightarrow$ MF   |
| EF $\nrightarrow$ GI         | 2.794*** (0.005)  | EF $\rightarrow$ GI    | MF $\nrightarrow$ GI         | 3.497 *** (0.000) | MF $\rightarrow$ GI    |
| GI $\nrightarrow$ EF         | 4.393*** (0.000)  | GI $\rightarrow$ EF    | GI $\nrightarrow$ MF         | 7.055*** (0.000)  | GI $\rightarrow$ MF    |
| EF $\nrightarrow$ lnGDP      | −0.223 (0.823)    |                        | MF $\nrightarrow$ lnGDP      | 0.294 (0.768)     |                        |
| lnGDP $\nrightarrow$ EF      | 8.638*** (0.000)  | lnGDP $\rightarrow$ EF | lnGDP $\nrightarrow$ MF      | 4.373 *** (0.000) | lnGDP $\rightarrow$ MF |
| EF $\nrightarrow$ RE         | 1.124 (0.260)     |                        | MF $\nrightarrow$ RE         | 2.899 *** (0.003) | MF $\rightarrow$ RE    |
| RE $\nrightarrow$ EF         | 12.393*** (0.000) | RE $\rightarrow$ EF    | RE $\nrightarrow$ MF         | 6.699*** (0.000)  | RE $\rightarrow$ MF    |

The symbols  $\rightarrow$ , and  $\nrightarrow$  specify a unidirectional causal link and no causality

\*\*\* $p < 0.01$

stimulates green innovation in material efficiency and recycling. Hence, MF pressure triggers GI. At the same time, GI, which is material-intensive, increases the MF. Regarding the bidirectional link between RE and MF, it can be explained as follows. First, the expansion of RE technologies requires the construction of plants, grids, and storage systems. This leads to an increase in the demand for construction minerals and critical metals. On the other hand, the level of MF, the availability of raw materials, and their costs influence the decisions towards the transition to RE.

## 5 Conclusion

EF is a multifaceted measure that includes factors such as the quality of water and land. Since this indicator evaluates pressure on the ecosystem but does not directly account for the consumption of raw materials, it is essential to jointly analyze the determinants of both MF and EF. Hence, this investigation focuses on the drivers of EF and MF in the G7. The panel CS-ARDL approach, together with a set of econometric tools, is used. Specifically, CSD and unit root tests are used. In addition, the second-generation cointegration test and the Granger non-causality test are also implemented.

The findings confirm that an increase in EPS is associated with lower EF. Hence, EPS is a major factor in protecting the environment in the G7 economies. These results confirm that stringent environmental strategies are very important in influencing the pace, trajectory, and environmental impacts of the green transition. Therefore, strict environmental policies influence the behavior of economic agents by motivating them to adopt RE and sustainable technologies in production and consumption, ultimately enhancing environmental conditions. However, our results suggest that rigorous ecological governance is not yet effective in promoting environmentally responsible consumption and production across all sectors. Therefore, environmental policies, which are generally limited to reducing pollutant emissions, must also be oriented towards the efficient use of raw materials to lower the MF.

Moreover, RE has a negative impact on EF. Hence, it is evident that the use of RE not only improves air quality by reducing carbon emissions but also negatively affects EF. It is recommended that the expansion of RE in the G7 economies be supported by energy efficiency policies and the adoption of smart grids. Such measures help reduce

losses in production, consumption, transmission, and distribution. Additionally, it is important to go beyond large-scale solar and wind projects and promote renewable energy solutions that are well integrated into local contexts, such as rooftop solar installations. Finally, the deployment of RE should be accompanied by stringent environmental policies to manage its full life-cycle impact, from the industrial production and use of critical minerals to the installation of large turbines and vast mirror fields, which can potentially disrupt ecosystems.

Furthermore, GI has a positive effect on MF. This suggests that the advancement of sustainable technologies in the G7 countries is currently contributing to an increase in the use of raw materials. Therefore, it is essential that these technologies address all dimensions of sustainability, not just air pollution. To move toward sustainability, it is crucial to use GI without destroying the environment through extensive extraction of resources. In this context, the application of strict environmental strategies concerning material extraction is a key measure. Such strategies will guarantee a gradual and sustainable green transition. For these reasons, policymakers must prioritize the circular economy, encourage resource-efficient green technologies, and ensure the existence of sustainable supply chains. For instance, setting mandatory standards concerning recycled content in green technologies is a good step toward adopting less material-intensive technologies.

Actually, this investigation is at the heart of the debate concerning the SDGs, particularly goals 7, 8 and 12, which aim to ensure access to sustainable energy, promote inclusive and sustainable economic growth, and foster sustainable consumption and production patterns. Particularly, investigating the determinants of MF is associated with SDG 12, and may help policymakers in their efforts to decouple growth from material-intensive activities.

Although this paper addresses a very important topic, it can be improved in both methodology and conceptualization. In fact, this investigation did not consider two interesting variables: trade and institutional quality. For future research, it would be useful to include the institutional quality variable and analyze its effect on the EF and MF, as was done in recent studies such as those of Mohamed et al. [38] and Wang et al. [57]. In particular, examining the interaction between EPS and institutional quality would be important, as many studies have shown that high institutional quality enhances the effectiveness of laws and measures. Moreover, Abid et al. [2] showed that improving institutional quality could enhance the sustainable exploitation of natural resources. In addition, it would be interesting to compare the consequences of the explanatory variables on other indicators, such as carbon emissions (e.g., [10]) or the load capacity factor (e.g., [41]), to provide a deeper analysis. Furthermore, since the CS-ARDL model fails to account for nonlinear effects, using nonlinear ARDL will be of great interest in future research. Finally, a comparison between the G7 and other groups of countries, such as the E7 or developing countries, will be very interesting once data are available for these panels.

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**Author contributions**

Emna Omri: Conceptualization; methodology; writing—original draft; writing—review and editing; validation; project administration; data curation; software; formal analysis. Chiraz Feki: Conceptualization; methodology; writing—original draft; writing—review and editing. Nouri Chtourou: Conceptualization; methodology; validation.

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Data will be made available on reasonable request.

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